

Rahil Ehtesham Mehr

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Education

Bachelor of Science in Physics

Sharif University of Technology, Tehran, Iran | 2022 - expected 2027

GPA: 15.1/20 (top third of class).

Research interests

Experimental condensed matter physics; Nanoscale materials and devices; Thin-film synthesis and characterization; Two-dimensional materials and heterostructures.

Research experience

Thin films and two-dimensional materials

Current research

Experimental research in thin films and two-dimensional materials, gaining hands-on experience in material fabrication and characterization.

Non-invasive laser-tissue interaction simulation

Started June 2024

Computational modeling of optical and thermal interactions between coherent light and biological tissue, with photon transport simulations in multilayer skin models.

Supervisor: Dr. Rasoul Sedighi

Academic projects

High-vacuum system design

Fall 2024

Design, optimization, and testing of a vacuum system for SEM and XPS instruments, with the aim of supporting nanoscale surface analysis.

Supervisor: Prof. Ali Esfandiar

Critical temperature in superconductors

Spring 2024

An investigation of methods for increasing the critical temperature of high-temperature superconductors and their potential practical applications.

Supervisor: Prof. Shant Baghran

Mechanical constraints and DNA persistence length

Spring 2025

Analysis of how confinement and tensile forces change the apparent persistence length of DNA, using the worm-like chain model as a reference.

Supervisor: Prof. Seyed-Nader Seyed Reihani

Mapping the universe through the cosmic microwave background

Spring 2023

An examination of how cosmic microwave background data reveal primordial density fluctuations and their connection to the universe we observe today.

Supervisor: Prof. Baharm Mashhoon and Farhad Ardalan

Honors & awards

University chess team

November 2022 - present

Sharif University of Technology. Second place with the team at the 2025 Iranian National Inter-University Chess Championship. Second place at the 2024 National University Sports Olympiad; first place individually and second place as a team at the 2024 Regional Tournament.

National University Entrance Exam (Concours)

Fall 2022

Ranked in the top 1% in Mathematics and Physics.

Professional chess player

2010-2020

Over 60 regional and national titles.

Teaching

Chess instructor

Shahmate chess school | March 2023 - July 2026

Certified chess instructor.

Chess instructor

Private group tutoring

Taught chess fundamentals to elementary students, developing age-appropriate lessons in tactics, strategy, and competitive play, with an emphasis on critical thinking and problem-solving.

Skills & languages

Programming and analysis: Python, Excel, LaTeX

Experimental materials research: Hands-on experience in thin-film and 2D-material fabrication and characterization

Characterization: Familiarity with SEM and XPS system operation

Research practice: Scientific report writing and team collaboration

Persian: Native

English: Professional working proficiency

Selected coursework

Course	Grade / 20	Course	Grade / 20
Vacuum Technology	17.7	Physics 4	18
Laser Applications	16.5	Physics 4 Lab	19.2
Electromagnetics	18	Introductory Programming	16.5
Biophysics	17	Philosophy and Society	18.5

Service

Chess puzzle columnist

Sharif University Newspaper | September 2025 - present

Contributes chess puzzles to the university newspaper.

Interdepartmental chess tournament director and organizer

Sharif University of Technology, Physics Department | May 2024 and May 2025

Managed event logistics and administration for tournaments involving faculty and students.